

ARTIFICIAL INTELLIGENCE

Lecture Set 15
K-Nearest Neighbor

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Course Contents

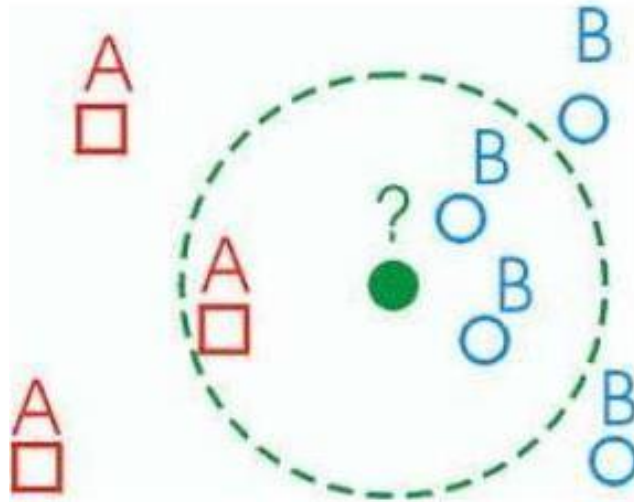
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11. Bayesian Networks, Naïve Bayes Classifier
12. Hidden Markov model, Markov Decision Process
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KNN

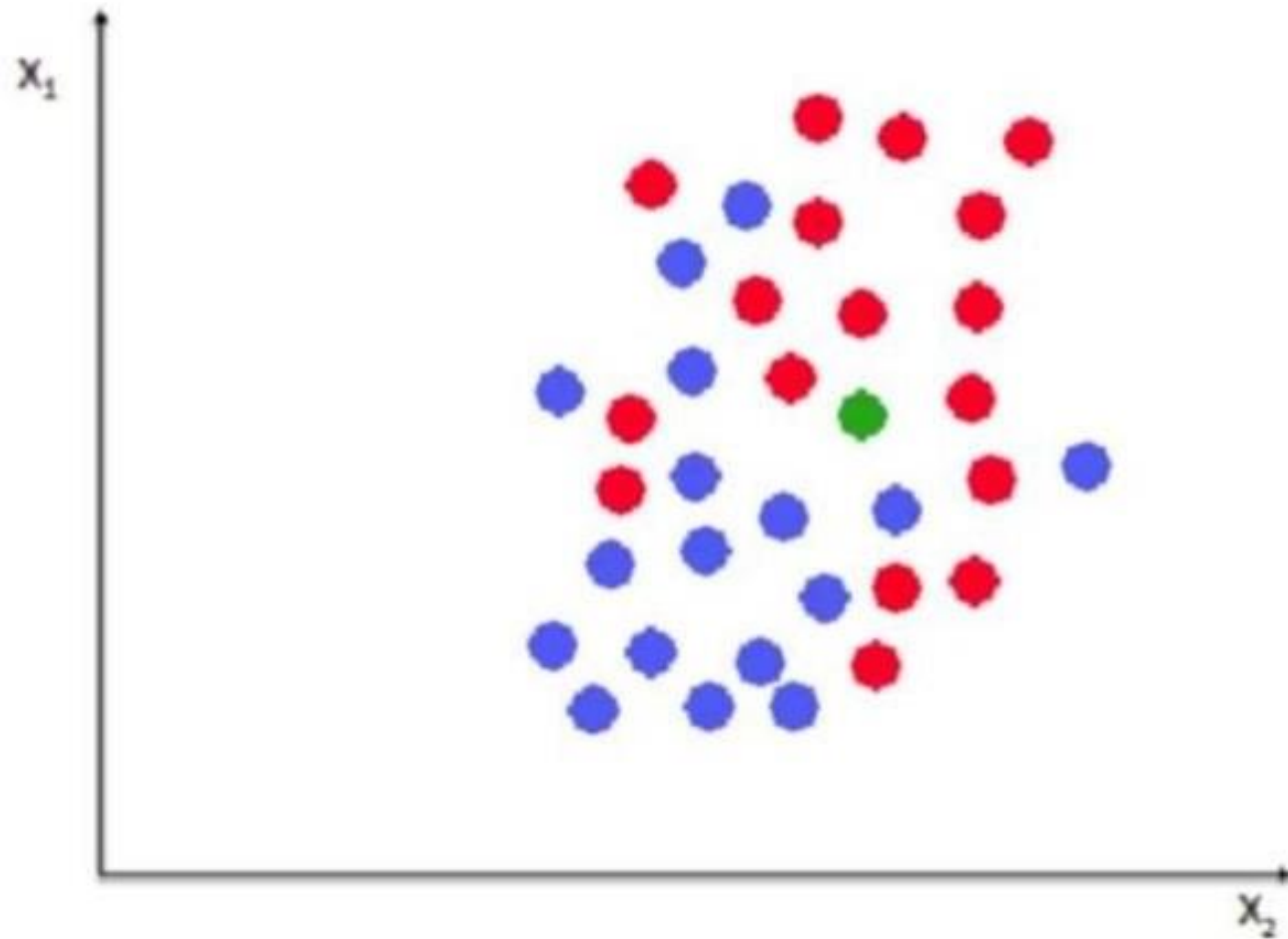
- K-Nearest Neighbors (KNN)
- Simple, but a very powerful classification algorithm
- Classifies based on a **similarity measure**
- Non-parametric
- Lazy learning
 - ▮ Does not “learn” until the test example is given
 - ▮ Whenever we have a new data to classify, we find its K-nearest neighbors from the training data

KNN: Classification Approach

- Classified by “**MAJORITY VOTES**” for its neighbor classes
- ▮ Assigned to the most common class amongst its K -nearest neighbors (by measuring “distance” between data)



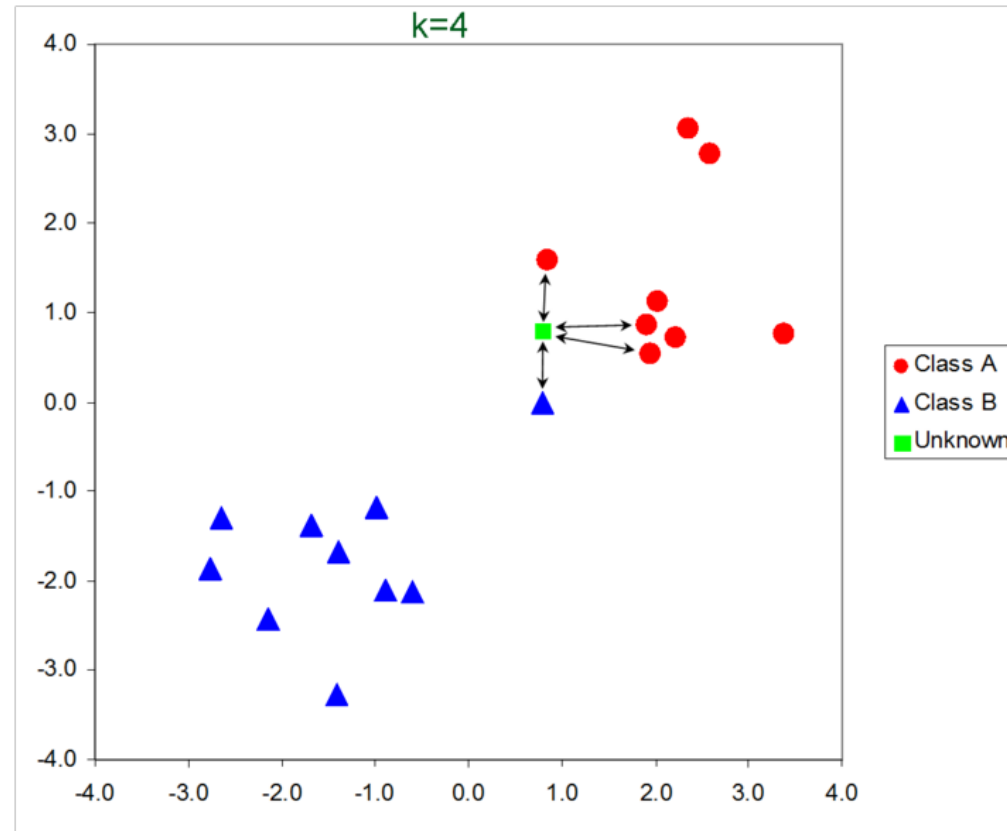
KNN: Example



KNN: Pseudocode

- Step 1: Determine parameter K = number of nearest neighbors
- Step 2: Calculate the distance between the query-instance and all the training examples.
- Step 3: Sort the distance and determine nearest neighbors based on the k -th minimum distance.
- Step 4: Gather the category Y of the nearest neighbors.
- Step 5: Use simple majority of the category of nearest neighbors as the prediction value of the query instance.

KNN: Example



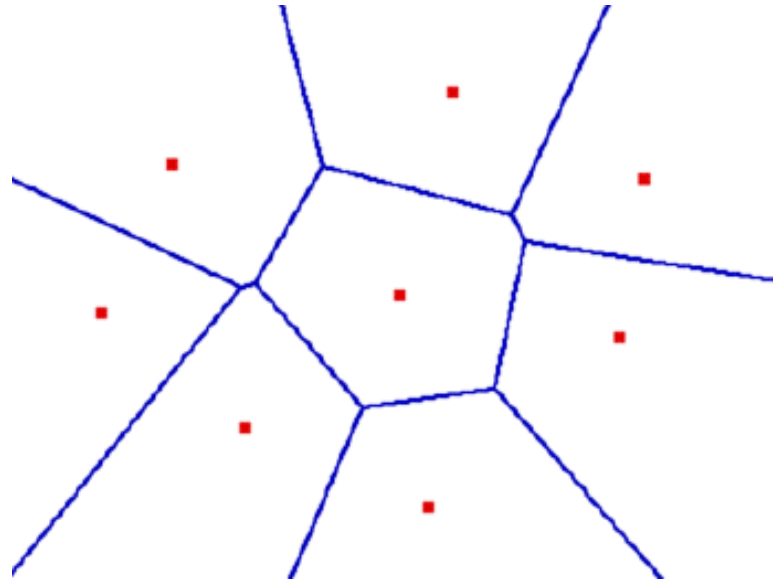
KNN: Euclidean distance matrix

Table 1. Euclidean distance matrix D listing all possible pairwise Euclidean distances between 19 samples.

	x ₁	x ₂	x ₃	x ₄	x ₅	x ₆	x ₇	x ₈	x ₉	x ₁₀	x ₁₁	x ₁₂	x ₁₃	x ₁₄	x ₁₅	x ₁₆	x ₁₇	x ₁₈
x ₂	1.5																	
x ₃	1.4	1.6																
x ₄	1.6	1.4	1.3															
x ₅	1.7	1.4	1.5	1.5														
x ₆	1.3	1.4	1.4	1.5	1.4													
x ₇	1.6	1.3	1.4	1.4	1.5	1.8												
x ₈	1.5	1.4	1.6	1.3	1.7	1.6	1.4											
x ₉	1.4	1.3	1.4	1.5	1.2	1.4	1.3	1.5										
x ₁₀	2.3	2.4	2.5	2.3	2.6	2.7	2.8	2.7	3.1									
x ₁₁	2.9	2.8	2.9	3.0	2.9	3.1	2.9	3.1	3.0	1.5								
x ₁₂	3.2	3.3	3.2	3.1	3.3	3.4	3.3	3.4	3.5	3.3	1.6							
x ₁₃	3.3	3.4	3.2	3.2	3.3	3.4	3.2	3.3	3.5	3.6	1.4	1.7						
x ₁₄	3.4	3.2	3.5	3.4	3.7	3.5	3.6	3.3	3.5	3.6	1.5	1.8	0.5					
x ₁₅	4.2	4.1	4.1	4.1	4.1	4.1	4.1	4.1	4.1	4.1	1.7	1.6	0.3	0.5				
x ₁₆	4.1	4.1	4.1	4.1	4.1	4.1	4.1	4.1	4.1	4.1	1.6	1.5	0.4	0.5	0.4			
x ₁₇	5.9	6.2	6.2	5.8	6.1	6.0	6.1	5.9	5.8	6.0	2.3	2.3	2.5	2.3	2.4	2.5		
x ₁₈	6.1	6.3	6.2	5.8	6.1	6.0	6.1	5.9	5.8	6.0	3.1	2.7	2.6	2.3	2.5	2.6	3.0	
x ₁₉	6.0	6.1	6.2	5.8	6.1	6.0	6.1	5.9	5.8	6.0	3.0	2.9	2.7	2.4	2.5	2.8	3.1	0.4

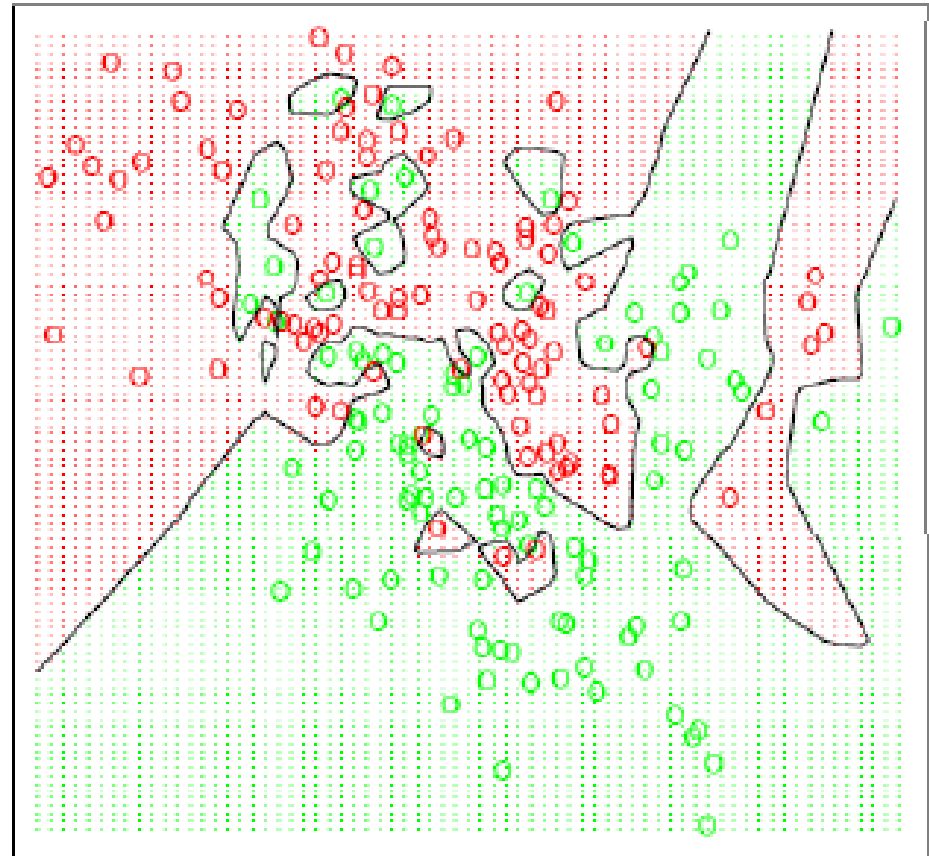
Decision Boundaries

- Voronoi diagram
 - ▮ Describes the areas that are nearest to any given point, given a set of data.
 - ▮ Each line segment is equidistant between two points of opposite class



Decision Boundaries

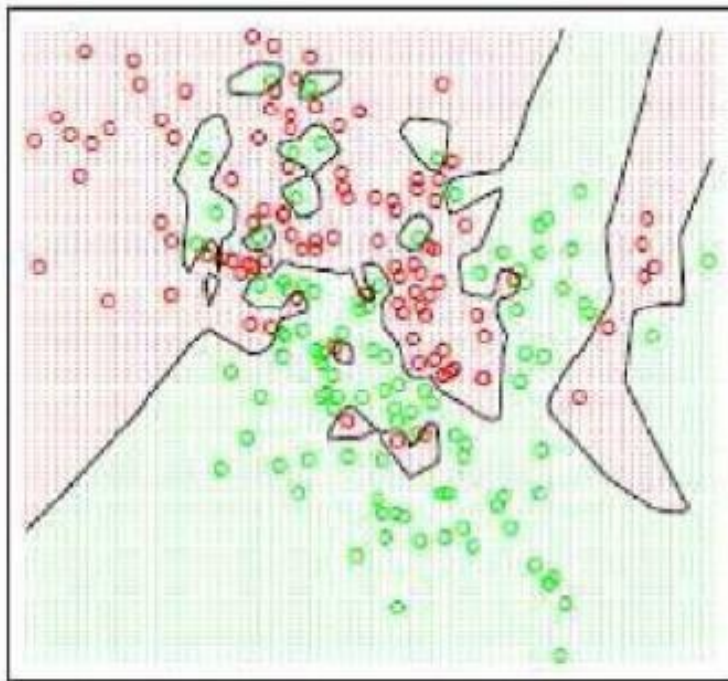
- With large number of examples and possible noise in the labels, the decision boundary can become nasty!
 - ▮ “Overfitting” problem



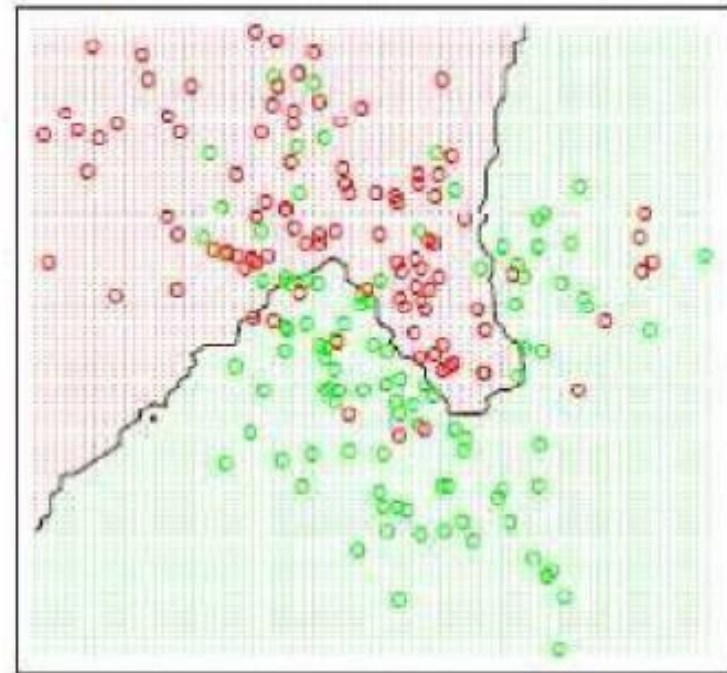
Effect of K

- Larger k produces smoother boundary effect
- When $K=N$, always predict the majority class

$K=1$



$K=15$



Figures from Hastie, Tibshirani and Friedman (Elements of Statistical Learning)

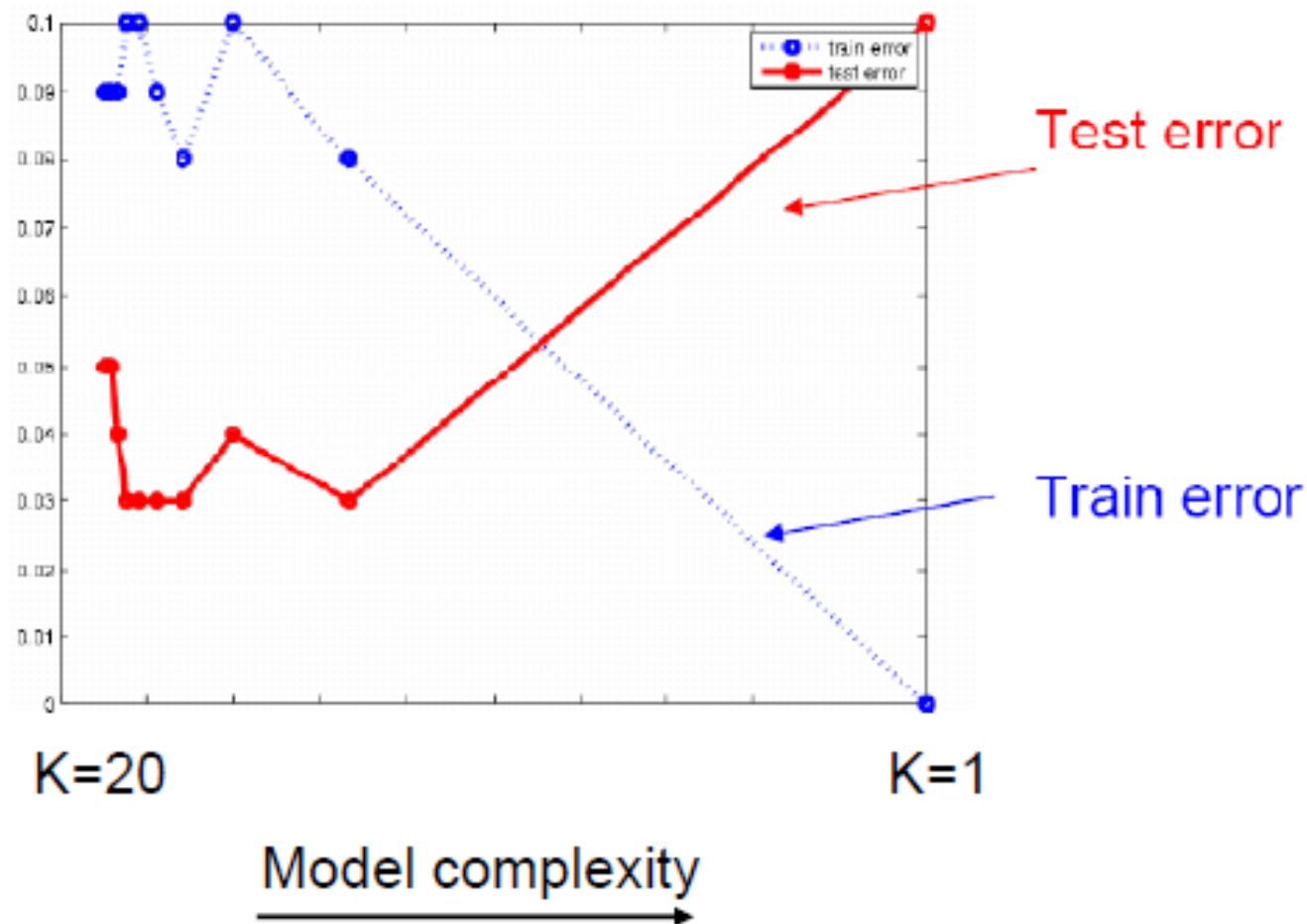
Discussion



- Which model is better between $K=1$ and $K=15$?
- Why?

How to choose k ?

□ Empirically optimal k ?



Pros and Cons



□ Pros

- ▮ Learning and implementation is extremely simple and Intuitive
- ▮ Flexible decision boundaries

□ Cons

- ▮ Irrelevant or correlated features have high impact and must be eliminated
- ▮ Typically difficult to handle high dimensionality
- ▮ Computational costs: memory and classification time computation

Similarity and Dissimilarity

□ Similarity

- ▮ Numerical measure of how alike two data objects are.
- ▮ Is higher when objects are more alike.
- ▮ Often falls in the range $[0, 1]$

□ Dissimilarity

- ▮ Numerical measure of how different are two data objects
- ▮ Lower when objects are more alike
- ▮ Minimum dissimilarity is often 0
- ▮ Upper limit varies

□ Proximity refers to a similarity or dissimilarity

Euclidean Distance

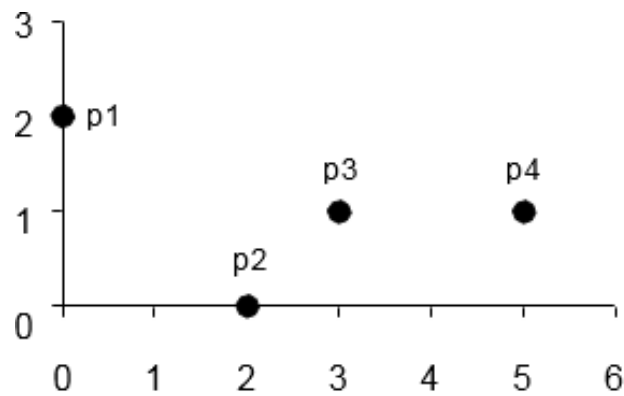
- Euclidean Distance

$$dist = \sqrt{\sum_{k=1}^p (a_k - b_k)^2}$$

Where p is the number of dimensions (attributes) and a_k and b_k are, respectively, the k -th attributes (components) or data objects a and b .

- Standardization is necessary, if scales differ.

Euclidean Distance



point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

Solved Example

	Maths	CS	Result
1)	4	3	Fail
2)	6	7	Pass
3)	7	8	Pass
4)	5	5	Fail
5)	8	8	Pass

Query $\rightarrow x = (\text{Maths}=6, \text{CS}=8)$
 $K=3$

- 1) Find distance
- 2) Find Rank
- 3) Find NN

Euclidean distance

$$d = \sqrt{|x_{01} - x_{A1}|^2 + |x_{02} - x_{A2}|^2}$$

$$1) \sqrt{(6-4)^2 + (8-3)^2} = \sqrt{29} = 5.38$$

$$2) \sqrt{(6-6)^2 + (8-7)^2} = \textcircled{1} \rightarrow \text{Pass}$$

$$3) \sqrt{(6-7)^2 + (8-8)^2} = \textcircled{1} \rightarrow \text{Pass}$$

$$4) \sqrt{(6-5)^2 + (8-5)^2} = \sqrt{10} = 3.16$$

$$5) \sqrt{(6-8)^2 + (8-8)^2} = \textcircled{2} \rightarrow \text{Pass}$$

$$PPP = 3P, OF$$

$$3P > OF$$

$$x \rightarrow (\text{Maths}=6, \text{CS}=8) \rightarrow \text{Pass}$$

Homework

BMI	Age	Sugar
33.6	50	1
26.6	30	0
23.4	40	0
43.1	67	0
35.3	23	1
35.9	67	1
36.7	45	1
25.7	46	0
23.3	29	0
31	56	1

- **Apply** K nearest neighbor classifier to predict the diabetic patient with the given features BMI, Age. If the training examples are,
- Assume K=3,
- Test Example BMI=43.6, Age=40, Sugar=?

Predicting Movie Genre

IMDb Rating	Duration	Genre
8.0 (Mission Impossible)	160	Action
6.2 (Gadar 2)	170	Action
7.2 (Rocky & Rani)	168	Comedy
8.2 (OMG 2)	155	Comedy

Apply KNN to find the genre of “Barbie” movie with IMDb rating 7.4 and duration 114 minutes when K=3.

Solved Example

$$d = \sqrt{(x_{o1} - x_{A1})^2 + (x_{o2} - x_{A2})^2}$$

$$x(\text{BMI}, \text{Age}) = (43.6, 40) ; K = 5$$

BMI	Age	Sugar	Distance	
33.6	50	1	$\sqrt{(43.6 - 33.6)^2 + (40 - 50)^2} = 14.14 \checkmark 2)$	1, 1
26.6	30	0	$\sqrt{(43.6 - 26.6)^2 + (40 - 30)^2} = 19.72 \checkmark 5)$	2), 1
23.4	40	0	$\sqrt{(43.6 - 23.4)^2 + (40 - 40)^2} = 20.20$	3), 0
43.1	67	0	$\sqrt{(43.6 - 43.1)^2 + (40 - 67)^2} = 27$	4), 1
35.3	23	1	$\sqrt{(43.6 - 35.3)^2 + (40 - 23)^2} = 18.92 \checkmark 4)$	5), 0
35.9	67	1	$\sqrt{(43.6 - 35.9)^2 + (40 - 67)^2} = 28.08$	3), 20s
36.7	45	1	$\sqrt{(43.6 - 36.7)^2 + (40 - 45)^2} = 8.52 \checkmark 1)$	3), > 20s
25.7	46	0	$\sqrt{(43.6 - 25.7)^2 + (40 - 46)^2} = 18.88 \checkmark 3)$	
23.3	29	0	$\sqrt{(43.6 - 23.3)^2 + (40 - 29)^2} = 23.09$	Therefore,
31	56	1	$\sqrt{(43.6 - 31)^2 + (40 - 56)^2} = 20.37$	$x(\text{BMI}, \text{Age}) \rightarrow 1$

Homework

Height (CM)	Weight (KG)	Class
167	51	Underweight
182	62	Normal
176	69	Normal
173	64	Normal
172	65	Normal
174	56	Underweight
169	58	Normal
173	57	Normal
170	55	Normal
170	57	?

Apply KNN to find the missing class labels when K=5.

Sepal Length	Sepal Width	Species
5.3	3.7	Setosa
5.1	3.8	Setosa
7.2	3.0	Virginica
5.4	3.4	Setosa
5.1	3.3	Setosa
5.4	3.9	Setosa
7.4	2.8	Virginica
6.1	2.8	Versicolor
7.3	2.9	Virginica
6.0	2.7	Versicolor
5.8	2.8	Virginica
6.3	2.3	Versicolor
5.1	2.5	Versicolor
6.3	2.5	Versicolor
5.5	2.4	?